JAIN (DEEMED-TO-BE UNIVERSITY)

PROJECT REPORT

COLLEGE INTELLIGENCE SLM

Small Language Model for College Information Question Answering

Submitted in partial fulfilment of the academic requirements

Submitted by	LEKIREDDY SREE PRIYA
Programme	B.Tech – Computer Science and Engineering (AI & ML)
Semester / Section	
Register Number	**25BTRCL128**
Faculty Guide	**SUJEET JAGTAP**
Academic Year	2026–2027

Department of Computer Science and Engineering
Artificial Intelligence and Machine Learning

Jain (Deemed-to-be University), Bengaluru

DOCUMENT INFORMATION

Item	Details
Project title	College Intelligence SLM – Small Language Model for College Information Question Answering
Base model	google/flan-t5-small
Implementation environment	Google Colab; NVIDIA Tesla T4 GPU (as recorded in notebook)
Model hosting	Hugging Face Model Hub
Interactive interface	Gradio (developed in Colab)
Report status	Academic project report prepared from the implementation details shared by the student

Personalization note: Complete the blank student, register number, semester/section and faculty-guide fields on the cover page before submission. Confirm the university's required cover-page template and citation style with the faculty guide.

ABSTRACT

This project presents College Intelligence SLM, a context-based question-answering system intended to answer questions about colleges, educational programmes and related institutional information. The implementation uses a fine-tuned version of the FLAN-T5-small sequence-to-sequence model. A structured college dataset containing 999 records and 72 columns was prepared, cleaned and transformed into 7,282 question-answer examples. The examples were divided into training (5,786), validation (416) and testing (1,080) sets. The notebook records an institution-group-based split and reports zero overlap between the groups assigned to the three splits.

The model was fine-tuned in Google Colab using an NVIDIA Tesla T4 GPU. The final training configuration recorded three epochs, batch size four, gradient accumulation of two and learning rate 0.00005. The notebook reports a final training loss of 0.001415 and validation loss of approximately 0.000004. On the prepared test set, the reported exact-match accuracy and average token F1 were both 100%. A Gradio chatbot interface was also created, and the saved model was uploaded to Hugging Face. A separate download-and-load check succeeded, followed by a corrected independent generation example that returned the expected institution name.

The reported evaluation values describe performance on the prepared dataset and should not be interpreted as proof of perfect performance on arbitrary real-world questions. Additional evaluation on genuinely unseen institutions, paraphrased questions and incomplete contexts is recommended.

TABLE OF CONTENTS

Section	Page
1. Introduction	3
2. Problem Definition and Objectives	4
3. Dataset Description and Preparation	5
4. System Design and Architecture	6
5. Model and Training Methodology	7
6. Experimental Setup and Training Results	8
7. Testing and Evaluation	9
8. Chatbot Implementation and Deployment	10
9. Results, Discussion and Limitations	11
10. Conclusion and Future Work	12
References	13
Appendix A – Reproduction and Submission Checklist	14

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
SLM	Small Language Model
NLP	Natural Language Processing
QA	Question Answering
GPU	Graphics Processing Unit
EM	Exact Match
F1	F1 score (token-level overlap metric)
JSON	JavaScript Object Notation
API	Application Programming Interface

1. INTRODUCTION

Educational information is often stored in structured files containing institutional names, programme titles, credentials and other attributes. Finding a precise answer from such data can require users to know the exact column names or navigate large tables. A question-answering interface can make this information easier to access by allowing users to ask questions in natural language.

College Intelligence SLM explores this approach using a small pretrained sequence-to-sequence language model. The system prepares question-answer examples from a college dataset, fine-tunes FLAN-T5-small and provides a chatbot interface for submitting questions. The published model is hosted on the Hugging Face Model Hub so that it can be downloaded independently of the original Colab training session.

1.1 Project Scope

The project focuses on context-based answers about information represented in the supplied college dataset. It does not claim to be a general-purpose assistant, an authoritative source for current admissions information, or a replacement for official institutional websites. Its answers depend on the quality, coverage and correctness of the source data and the context supplied to the model.

1.2 Motivation

A compact model can be useful for demonstrating a complete applied AI workflow: dataset inspection, cleaning, example construction, group-aware data splitting, fine-tuning, evaluation, user-interface development and model publication. The project also provides a practical example of using an existing pretrained model rather than training a language model from scratch.

1.3 Report Organization

The report describes the problem and objectives, dataset preparation, architecture, model training, evaluation, chatbot interface, deployment, limitations and possible future improvements. Implementation values are reproduced from the progress and outputs shared during development; items not verified from the notebook are identified as recommendations or fields to confirm.

2. PROBLEM DEFINITION AND OBJECTIVES

2.1 Problem Statement

The project addresses the task of answering natural-language questions about college and programme records stored in a structured dataset. The system should convert relevant records into a usable context, accept a user question and generate an answer grounded in that context. It should avoid guessing when the supplied information does not support an answer.

2.2 Objectives

- Inspect and prepare a structured college-information dataset.
- Generate question-answer training examples from available record fields.
- Separate training, validation and test examples while avoiding institution-group overlap.
- Fine-tune the FLAN-T5-small pretrained model using the training examples.
- Monitor training and validation loss and evaluate the held-out test set.
- Build an interactive Gradio chatbot interface.
- Publish the model files and documentation to Hugging Face.
- Store the source notebook and project documentation in a GitHub repository.

2.3 Expected Outcome

The expected outcome is a reproducible educational prototype consisting of a prepared QA dataset, a fine-tuned model, evaluation outputs, an interactive chatbot and public project repositories. The Hugging Face repository is sree02200/college-intelligence-slm. The GitHub repository shared during development is College-Intelligence-SLM.

3. DATASET DESCRIPTION AND PREPARATION

3.1 Original Dataset

The supplied original JSON dataset was uploaded to Google Colab as `/content/dataset.json`. The dataset inspection output reported 999 rows and 72 columns. The data included institutional and programme-related attributes. The notebook output also reported 103 unique institutions and 20,941 missing values during the cleaning stage.

Dataset property	Recorded value
Original records	999
Columns	72
Unique institutions reported	103
Missing values reported during cleaning	20,941
QA examples generated	7,282

3.2 Data Cleaning and Preparation

The notebook included a data-preparation stage that restored/derived an institution grouping field named `group_id`. The output showed an example institution, “A - Technical College”, and an example programme, “Health and Medical Administrative Services”. Missing values were inspected and relevant record fields were used to build question-answer examples.

3.3 Question-Answer Example Generation

The notebook generated 7,282 QA examples. The displayed examples included questions such as “Which institution offers this program?”, “What is the name of this program?” and “What credential does this program offer?”. Each generated example included a question, an input text containing an instruction and context, and a target text containing the expected answer.

3.4 Context-Support Audit

An initial audit reported that 3,019 answers were present in the generated context and 4,263 were not present, corresponding to 41.46% context support. A correction stage was then run, after which the notebook reported 7,282 supported answers and 100.0% context support. This correction is an important part of the data pipeline and should be reviewed against the original dataset to ensure the target answers are truly supported by the input context and are not introduced by a transformation error.

3.5 Dataset Splitting

The recorded split contained 5,786 training examples, 416 validation examples and 1,080 testing examples. The notebook reported an institution-group leakage check with zero overlap for train/validation, train/test and validation/test. Group-based separation is intended to test generalization to institution groups not used for training. The exact splitting code and random seed should be retained in the submitted notebook for reproducibility.

Subset	Examples	Approx. share
Training	5,786	79.5%
Validation	416	5.7%
Testing	1,080	14.8%

Subset	Examples	Approx. share
Total	7,282	100%

Note: Percentages are rounded. The reported group separation is based on the notebook's own leakage-check output.

4. SYSTEM DESIGN AND ARCHITECTURE

4.1 High-Level Architecture

The system is organized as a pipeline. The structured dataset is inspected and cleaned, QA examples are generated, and the examples are separated into training, validation and test subsets. The training subset is used to fine-tune FLAN-T5-small. The resulting model is evaluated, saved and uploaded to Hugging Face. A chatbot interface uses the model to generate answers to user questions.

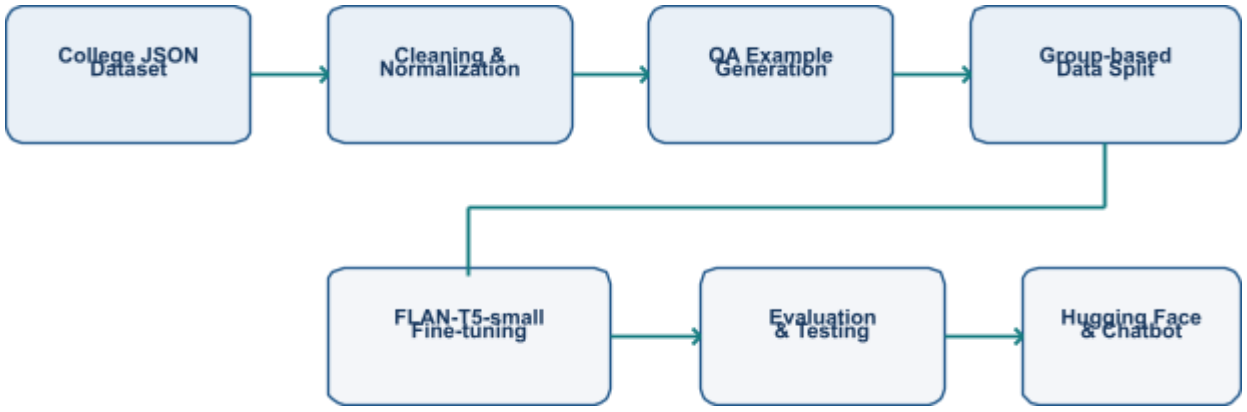


Figure 1. High-level workflow of the College Intelligence SLM project.

4.2 Main Components

Component	Purpose
Dataset loader	Reads the uploaded JSON file into a data structure for inspection and preparation.
Preprocessing	Cleans fields and derives an institution grouping identifier.
QA generator	Creates question, instruction/context input and target-answer examples.
Data splitter	Creates training, validation and test sets and checks institution-group overlap.
Tokenizer	Converts input and target text into token IDs suitable for FLAN-T5.
Fine-tuned model	Generates answer text from the tokenized input context.
Evaluation	Measures loss, exact match and token F1 on held-out examples.
Chatbot / deployment	Provides a Gradio interface and publishes the saved model on Hugging Face.

4.3 Model Input and Output

The model is a text-to-text sequence-to-sequence model. The input is an instruction containing the context and the question. The target is the expected answer string. At inference time, the input is tokenized, passed to the model's generation method and decoded into readable text. The quality of the answer therefore depends on the information included in the input context and on how the training examples represent the task.

4.4 Technology Stack

The implementation used Python in Google Colab, PyTorch for model computation, Hugging Face Transformers for model and tokenizer loading, the Datasets library for dataset mapping, Pandas for tabular data handling, Scikit-learn for group-aware splitting, Gradio for the chatbot interface and the Hugging Face Hub client for publishing the model.

5. MODEL AND TRAINING METHODOLOGY

5.1 Base Model: FLAN-T5-small

The project uses **google/flan-t5-small** as its pretrained base model. FLAN-T5 is an encoder-decoder Transformer model trained to follow instructions across text-to-text tasks. The small variant is suitable for an educational fine-tuning experiment where computational resources are limited. This project fine-tunes the pretrained model for the specific college-information question-answering format.

5.2 Tokenization

The tokenizer converts input prompts and expected answers into integer token sequences. The notebook output showed tokenization completed for all three subsets: 5,786 training, 416 validation and 1,080 test examples. The input and target sequences are padded/truncated according to the configured maximum lengths in the notebook. The exact maximum sequence lengths should be checked in the source notebook and reported consistently if the faculty requires them.

5.3 Fine-Tuning Configuration

Parameter	Final configuration recorded
Pretrained model	google/flan-t5-small
Epochs	3
Per-device batch size	4
Gradient accumulation steps	2
Learning rate	0.00005
Mixed precision	FP16: False
GPU	NVIDIA Tesla T4
Optimizer / scheduler	Confirm from submitted notebook training arguments

5.4 Training Procedure

The fine-tuning procedure loads the pretrained tokenizer and sequence-to-sequence model, tokenizes the prepared training and validation examples, and uses the training framework to optimize model parameters. Validation loss is monitored at the end of each epoch. The final model and tokenizer are saved to `/content/college_slm_verified`. The notebook also saves evaluation metrics and test predictions in the model output directory.

5.5 Numerical Stability Correction

An earlier training attempt reported zero training loss and NaN validation loss. A diagnostic check showed that the test loss was not finite. The notebook then reloaded the original FLAN-T5 model and confirmed that a fresh model produced a finite test loss. A corrected training configuration was subsequently used, with batch size four, gradient accumulation two, learning rate 0.00005 and FP16 disabled. The later training run reported finite training and validation losses. This correction and the earlier failure should be retained in the project history as part of the implementation record.

5.6 Saved Artifacts

The saved model folder contained README.md, config.json, evaluation_metrics.json, generation_config.json, model.safetensors, test_predictions.csv, tokenizer.json, tokenizer_config.json and training_args.bin. The Hugging Face upload excluded test_predictions.csv. The published model was subsequently downloaded and loaded successfully using the Hugging Face Transformers library.

6. EXPERIMENTAL SETUP AND TRAINING RESULTS

6.1 Experimental Environment

Item	Recorded environment
Platform	Google Colab
Accelerator	NVIDIA Tesla T4
Deep learning framework	PyTorch
Model library	Hugging Face Transformers
Base model	google/flan-t5-small
Training examples	5,786
Validation examples	416

6.2 Training Loss History

The final corrected training run reported the following losses across three epochs. The values below are reproduced from the training output shared during development.

Epoch	Training loss	Validation loss
1	0.015408	0.000011
2	0.002995	0.000005
3	0.001415	0.000004

Figure 2. Training and validation loss values from the final corrected training run.

The recorded training loss decreased across the three epochs. Validation loss also decreased slightly across the same epochs. The notebook later reported a final validation loss of approximately 4.156789145781659 × 10■■. Loss values depend on the training configuration, token masking and aggregation method; they should be interpreted together with held-out test metrics and qualitative examples.

6.3 Training Completion

The final training output displayed completion at epoch 3 of 3 and saved the model shards. The notebook printed a training runtime of approximately 950.53 seconds for the final run. The exact runtime can vary with Colab hardware availability and should not be treated as a fixed requirement.

6.4 Model Saving

The final model was saved locally at `/content/college_slm_verified` and archived as `/content/college_slm_verified.zip`. The Hugging Face repository was created as `sree02200/college-intelligence-slm`. The upload completed successfully, and the model card was updated with project metadata and documentation.

7. TESTING AND EVALUATION

7.1 Evaluation Dataset

The notebook evaluated 1,080 test examples. The test set was held out from training and validation using institution groups. The notebook reported zero institution-group overlap among the three subsets.

7.2 Evaluation Metrics

Exact Match (EM): measures the proportion of predictions that match the reference answer exactly after the notebook's normalization procedure. **Token F1:** measures token-level overlap between the generated answer and the reference answer using precision and recall. The exact implementation and normalization rules should be preserved in the notebook for reproducibility.

Metric	Reported result	Interpretation
Test examples evaluated	1,080	Number of held-out QA examples evaluated.
Exact Match Accuracy	100.0%	All evaluated predictions matched under the implemented normalization.
Average Token F1	100.0%	All evaluated predictions achieved full token overlap under the implemented metric.

7.3 Interpretation and Caution

The reported 100% values are unusually high and should be presented carefully. They may reflect the structured nature of the dataset, the way QA examples were generated from the same source records, repeated answer patterns, or the evaluation and normalization procedure. Group-based splitting reduces one form of leakage, but it does not by itself prove that the test questions are independent in every meaningful way. The dataset-generation code, answer-support correction and metric implementation should be audited before claiming broad generalization.

7.4 Independent Model Loading Test

After uploading the model to Hugging Face, the project downloaded the repository using AutoTokenizer.from_pretrained and AutoModelForSeq2SeqLM.from_pretrained. The output confirmed that the tokenizer and model loaded successfully and that the model was placed on the CUDA device in the active Colab session. This verifies that the published repository contains files that can be loaded through Transformers.

7.5 Independent Generation Example

An initial independent generation example returned a program identifier rather than the institution name. The input context was then reordered so that the institution field appeared first. The corrected example generated "A Better U Beauty Barber Academy", matching the expected answer. This was one manually selected example and is not a substitute for a comprehensive independent evaluation.

8. CHATBOT IMPLEMENTATION AND DEPLOYMENT

8.1 Gradio Interface

A chatbot interface was created using Gradio in Google Colab. The interface accepts a natural-language question and displays the generated answer. Example prompts shown during development included questions about the institution offering a programme, the programme name and median student debt. The Gradio interface was launched with a temporary public share URL during the development session.

8.2 Inference Workflow

The chatbot inference workflow consists of receiving a question, preparing the prompt and context, tokenizing the input, invoking the model's generate method and decoding the output. A robust implementation should retrieve the relevant college record before generation and include only the fields needed to answer the question. If no supporting record is found, the system should state that the information is unavailable rather than inventing an answer.

8.3 Hugging Face Deployment

A Hugging Face account was created and authenticated from Colab using a Write access token stored in Google Colab Secrets under the name HF_TOKEN. The model repository was created with the ID `sree02200/college-intelligence-slm`. The model folder was uploaded using the Hugging Face Hub API. The README model card was updated to describe the model, dataset, training setup, evaluation and limitations.

8.4 Published Repository

Hugging Face model: https://huggingface.co/sree02200/college-intelligence-slm
GitHub source repository: https://github.com/lekireddysreepriya247-alt/College-Intelligence-SLM

8.5 Reproducibility and Security

The source notebook should be committed to GitHub after removing tokens, private credentials and any sensitive data. The Hugging Face token must remain in Colab Secrets or another secure secret store and must never be committed to the repository. Large model weights should remain on Hugging Face rather than being uploaded directly to a standard GitHub repository. The dataset should only be redistributed if its licence and source terms permit it.

9. RESULTS, DISCUSSION AND LIMITATIONS

9.1 Achievements

- A structured dataset of 999 records and 72 columns was inspected and prepared.
- 7,282 QA examples were generated and divided into training, validation and testing subsets.
- The notebook reported zero overlap between institution groups across the three subsets.
- A corrected fine-tuning run completed for three epochs with finite loss values.
- The model, tokenizer and supporting configuration files were saved.
- The model was uploaded to Hugging Face and subsequently downloaded and loaded successfully.
- A Gradio chatbot interface was developed, and a corrected independent example returned the expected institution name.

9.2 Limitations

- The model's answers depend on the quality and coverage of the supplied college dataset.
- The project's reported 100% test metrics require independent auditing and should not be generalized to all user questions.
- Only one manually corrected independent generation example is documented in the development evidence.
- Questions requiring current information may receive outdated answers if the underlying dataset is not updated.
- The chatbot may produce an incorrect answer if the wrong record or incomplete context is supplied.
- The temporary Gradio share URL is not a permanent production deployment.
- The original dataset's redistribution permission and licence should be confirmed before making it public.

9.3 Recommended Additional Validation

- Evaluate a random sample of test predictions manually and inspect mismatches.
- Create paraphrased questions that were not used during QA generation.
- Test institutions that are entirely absent from the training data.
- Test missing-field, ambiguous and unsupported questions.
- Compare model-generated answers with a deterministic lookup baseline for structured fields.
- Report exact-match normalization rules and token-F1 implementation details.
- Verify that the context supplied at inference contains the correct record and is not truncated.

9.4 Ethical and Responsible Use

The system should be treated as an educational prototype. Users should verify important admissions, fee, accreditation and programme details with the relevant institution. The chatbot should not fabricate missing facts or present generated responses as official university statements. Any public release should respect the dataset owner's terms and avoid exposing personal or confidential information.

10. CONCLUSION AND FUTURE WORK

10.1 Conclusion

College Intelligence SLM demonstrates an end-to-end workflow for adapting a pretrained small language model to a focused educational question-answering task. The project covers dataset inspection and preparation, QA example generation, group-based splitting, tokenizer preparation, FLAN-T5-small fine-tuning, model evaluation, chatbot development and publication to Hugging Face. The final corrected training run completed on a Tesla T4 GPU in Google Colab, and the saved model was successfully downloaded and loaded from its Hugging Face repository.

The notebook reported 100% exact-match accuracy and 100% average token F1 on 1,080 prepared test examples. These results are encouraging for the prepared dataset but require further independent validation before making claims about general performance. The project therefore provides a working academic prototype and a foundation for further improvement.

10.2 Future Work

- Improve retrieval so that the model receives the most relevant institution record for each question.
- Expand the dataset with more institutions, programmes and carefully reviewed question paraphrases.
- Perform more extensive manual evaluation and error analysis.
- Add an explicit abstention mechanism for unsupported questions.
- Measure retrieval accuracy separately from answer-generation accuracy.
- Build a permanent hosted interface with suitable access controls and monitoring.
- Document data provenance, licence and update procedures.

10.3 Final Deliverables

Deliverable	Location / reference
Source notebook	GitHub repository – notebooks folder (to be organized)
Project report	This PDF report
Fine-tuned model	Hugging Face: sree02200/college-intelligence-slm
Chatbot interface	Gradio prototype created in Google Colab; temporary link may expire
Dataset	Original JSON used in Colab; publish only if redistribution is permitted

REFERENCES

[1] Google, “FLAN-T5-small,” Hugging Face model repository. Available: https://huggingface.co/google/flan-t5-small. Accessed 13 November 2026.

[2] Hugging Face, “Transformers Documentation.” Available: https://huggingface.co/docs/transformers.

[3] Hugging Face, “Hugging Face Hub Documentation.” Available: https://huggingface.co/docs/huggingface_hub.

[4] PyTorch, “PyTorch Documentation.” Available: https://pytorch.org/docs/stable/index.html.

[5] Gradio, “Gradio Documentation.” Available: https://www.gradio.app/docs.

[6] Scikit-learn, “GroupShuffleSplit Documentation.” Available: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GroupShuffleSplit.html.

[7] Project implementation notebook and Colab execution outputs shared during development. Dataset counts, training settings, evaluation metrics and deployment details in this report are based on those outputs.

Note: Add any dataset source citation, course-specific references and any additional libraries actually used in the final notebook. Confirm the required citation format with the faculty guide.

APPENDIX A – REPRODUCTION AND SUBMISSION CHECKLIST

A.1 Source Code Checklist

- Download the complete Google Colab notebook as an .ipynb file.
- Remove access tokens, private keys and credentials from notebook cells and outputs.
- Check that all paths and required files are documented.
- Include the final corrected training and evaluation cells, not only earlier failed attempts.
- Add a requirements.txt file listing the dependencies used.

A.2 GitHub Checklist

- Upload the notebook to the notebooks/ folder.
- Upload this PDF report to the reports/ folder.
- Add a README.md with the project summary, setup instructions and Hugging Face link.
- Add a .gitignore file for secrets, virtual environments, checkpoints and large model files.
- Do not upload the Hugging Face access token.
- Publish dataset files only when sharing is permitted.

A.3 Hugging Face Reference

Model repository: https://huggingface.co/sree02200/college-intelligence-slm
GitHub repository: https://github.com/lekireddysreepriya247-alt/College-Intelligence-SLM

A.4 Items to Complete Before Submission

Complete the student name, register number, semester/section and faculty-guide details on the cover page. Confirm whether the university requires a certificate, declaration, acknowledgement, plagiarism statement, specific logo, margins or binding format. Add actual screenshots from the notebook and chatbot if required by the faculty. Verify the repository is accessible to the faculty and that all uploaded files open correctly.

End of Report